

RFID Authenticated User Manager

Project Members

- Elias Reiman Treutiger (051127-3179)
- Erik Forsberg (990430-5852)

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Objective and Requirements

The purpose of this project is to develop an RFID database manager on an embedded platform with limited resources. The manager should be able to add, remove and edit users along with their associated RFID's by scanning tags. The main requirements for the manager are as follows:

- In-memory database.
- RFID authenticated user retrieval.
- Adding / removing users and their associated RFID tag to / from database.
- Editable per-user permissions.
- Editable user names.
- Graphical user interface.

Optional features that may be included, if time allows:

- Database encryption with part of RFID as key.
- Buzzer/Speaker.
- Non-volatile storage.

Solution.

We intend to develop our project on the DTEK-V platform together with an RFID-reader of the RC522 model, connected to the DTEK-V board's GPIO port via jumper cables. The user interface will be displayed on a 2x16 character LCD display also connected to the GPIO port via jumper cables. The embedded push buttons will be used to navigate the user interface options. A timer and interrupt will be incorporated in the implementation of the RFID reading functionality. The project will be developed in the C programming language.

Verification.

We will perform isolated tests of the different parts of the project before combining them into one full system. The parts to test are:

- The RFID reader, ensuring the received signals are correct and consistent.
- The database, making sure retrieval and modification of data is fast and correct, and that the kept RFID records are unique to avoid collisions.
- The screen and user interface, ensuring easy navigation and correct data display.
- If implemented, cryptography testing to ensure symmetric encryption works

Contributions.

We intend to divide the work equally, working together on every part as much as possible and only delegating tasks if necessary due to time constraints.

Reflections

In the final abstract, we will discuss and reflect on what happened in the project.